

2. Introducing probability

Recall: - sample space Ω

- set of events (σ -algebra) $\mathcal{F} \subseteq \mathcal{P}(\Omega)$

• $\Omega \in \mathcal{F}$

• $E \in \mathcal{F} \Rightarrow E^c = \Omega \setminus E \in \mathcal{F}$

• $E_1, E_2, \dots \in \mathcal{F} \Rightarrow \bigcup_i E_i \in \mathcal{F}$

- $E_1, E_2, \dots \in \mathcal{F}$ are mutually exclusive, if $\forall i \neq j: E_i \cap E_j = \emptyset$

- $E_1, E_2, \dots \in \mathcal{F}$ is a partition / complete system of events if $\forall i \neq j: E_i \cap E_j = \emptyset$ and $\bigcup_i E_i = \Omega$.

- Example: Roll a dice

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad \mathcal{F} = \mathcal{P}(\Omega) = \{\{1\}, \dots, \{6\}, \{1, 2\}, \{1, 3\}, \dots\}$$

$$\left. \begin{array}{l} E_1 = \{1, 3, 5\} \\ E_2 = \{2, 4, 6\} \end{array} \right\} \text{It's a partition.}$$

Note: $\mathcal{F}' = \{\emptyset, E_1, E_2, \Omega\}$ is also a σ -algebra.

1. Properties of probability

def. $P: \mathcal{F} \rightarrow \mathbb{R}$ is called probability (measure) if

$\forall E \in \mathcal{F} \Rightarrow P(E) \geq 0$

$\forall E_1, E_2, \dots \in \mathcal{F}$ mutually exclusive $\Rightarrow P(\bigcup_i E_i) = \sum_i P(E_i)$

$\Rightarrow P(\Omega) = 1$

Prop. $\forall E \in \mathcal{F}: P(E^c) = 1 - P(E)$

proof: $\Omega = E \cup E^c$ and $E \cap E^c = \emptyset \Rightarrow$ they are mutually exclusive.

$$1 = P(\Omega) = P(E \cup E^c) \stackrel{\text{me.}}{=} P(E) + P(E^c) \Rightarrow P(E^c) = 1 - P(E) \quad \square$$

Cor. $\forall E \in \mathcal{F}: 0 \leq P(E) \leq 1$

proof: $E \in \mathcal{F} \Rightarrow E^c \in \mathcal{F} \Rightarrow P(E^c) = 1 - P(E) \geq 0 \Rightarrow P(E) \leq 1 \quad \square$

Cor. $P(\emptyset) = 0$

Thm. $\forall E, F \in \mathcal{F} \Rightarrow P(E \cup F) = P(E) + P(F) - P(E \cap F)$

proof: $P(E \cup F) = P(F \cup (E \setminus F)) \stackrel{\text{me.}}{=} P(F) + P(E \setminus F) \stackrel{\text{me.}}{=} P(F) + P(E) - P(E \cap F)$

$$E = (E \cap F) \cup (E \setminus F) \Rightarrow P(E) \stackrel{\text{me.}}{=} P(E \cap F) + P(E \setminus F)$$

Cor.1 Let $E, F \in \mathcal{F}$ such that $E \subseteq F$, then

$$a) P(F \setminus E) = P(F) - P(E)$$

$$b) P(E) \leq P(F)$$

proof: a) Using again that $F = \underbrace{(E \cap F)}_{=E} \cup (F \setminus E)$

$$\Rightarrow P(F) = P(E) + P(F \setminus E)$$

$$b) F \setminus E = F \cap E^c = (F^c \cup E)^c \Rightarrow F \setminus E \in \mathcal{F}$$

$$\Rightarrow 0 \leq P(F \setminus E) = P(F) - P(E) \Rightarrow P(E) \leq P(F) \quad \square$$

Thm.1 (Continuity of the probability)

Let $E_1, E_2, \dots \in \mathcal{F}$ be a countable sequence of events such that $\forall n \geq 1: E_n \subseteq E_{n+1}$, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n).$$

proof: We define the following sequence of events

$$F_1 = E_1, F_2 = E_2 \setminus E_1, \dots, F_n = E_n \setminus E_{n-1}, \dots$$

$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n \Rightarrow P\left(\bigcup_{n=1}^{\infty} E_n\right) = P\left(\bigcup_{n=1}^{\infty} F_n\right) \underset{\text{me.}}{=} \sum_{n=1}^{\infty} P(F_n) =$$

$$\underset{\text{by definition of } (F_n)_{n \geq 1}}{=} \lim_{n \rightarrow \infty} \sum_{k=1}^n P(F_k) \underset{\text{me.}}{=} \lim_{n \rightarrow \infty} P\left(\bigcup_{k=1}^n F_k\right) = \lim_{n \rightarrow \infty} P(E_n) \quad \square$$

Cor.1 (Continuity of the probability)

Let $E_1, E_2, \dots \in \mathcal{F}$ be a countable sequence of events such that $\forall n \geq 1: E_n \supseteq E_{n+1}$, then

$$P\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n).$$

proof:

$$P\left(\bigcap_{n=1}^{\infty} E_n\right) = 1 - P\left(\left(\bigcap_{n=1}^{\infty} E_n\right)^c\right) \underset{\text{De Morgan}}{=} 1 - P\left(\bigcup_{n=1}^{\infty} E_n^c\right) =$$

$$= 1 - \lim_{n \rightarrow \infty} P(E_n^c) = 1 - \lim_{n \rightarrow \infty} (1 - P(E_n)) = \lim_{n \rightarrow \infty} P(E_n) \quad \square$$

$E_n^c \subseteq E_{n+1}^c$, so it follows from the theorem

Prop.1 $E_1, E_2, \dots \in \mathcal{F} \Rightarrow P\left(\bigcup_i E_i\right) \leq \sum_i P(E_i)$

proof: We first show the statement for finitely many sets, using mathematical induction.

I) It works for 2 sets:

$$E_1, E_2 \in \mathcal{F} \Rightarrow P(E_1 \cup E_2) \stackrel{\text{thm.}}{=} P(E_1) + P(E_2) - \underbrace{P(E_1 \cap E_2)}_{0 \leq} \leq P(E_1) + P(E_2)$$

since $(E_1 \cap E_2)^c = E_1^c \cup E_2^c \in \mathcal{F}$

II) We assume it works for n sets!

III) Using the assumption we show it works for $(n+1)$ sets:

$$E_1, E_2, \dots, E_n, E_{n+1} \in \mathcal{F}$$

$$P(\underbrace{E_1 \cup \dots \cup E_n}_{\mathcal{F}} \cup E_{n+1}) \stackrel{\text{it works for 2 sets}}{\leq} P(\mathcal{F}) + P(E_{n+1}) \stackrel{\text{it works for } n \text{ sets}}{\leq} P(E_1) + \dots + P(E_n) + P(E_{n+1})$$

$P(\mathcal{F}) \leq P(E_1) + \dots + P(E_n)$

Now we prove the statement for countably many sets, with the help of the continuity of the probability.

$$E_1, E_2, \dots \in \mathcal{F} \text{ and let } G_k = \bigcup_{i=1}^k E_i \Rightarrow G_k \subseteq G_{k+1} \text{ and } \bigcup_{k=1}^{\infty} G_k = \bigcup_{k=1}^{\infty} E_k$$

$$\text{Thus } P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{k=1}^{\infty} G_k\right) \stackrel{\text{continuity}}{=} \lim_{k \rightarrow \infty} P(G_k) = \lim_{k \rightarrow \infty} P\left(\bigcup_{i=1}^k E_i\right) \leq \lim_{k \rightarrow \infty} \sum_{i=1}^k P(E_i) = \sum_{i=1}^{\infty} P(E_i)$$

we already showed it for finite unions □

Thm: (Sieve formula / Inclusion-exclusion)

$$\forall E_1, \dots, E_n \in \mathcal{F} \Rightarrow P(E_1 \cup \dots \cup E_n) = \sum_{k=1}^n \sum_{1 \leq i_1 < \dots < i_k \leq n} (-1)^{k+1} P(E_{i_1} \cap \dots \cap E_{i_k})$$

proof: We prove it by mathematical induction.

I) We already proved it for 2 events. (earlier theorem)

II) Assume it works for n events!

III) Using the assumption we show it works for $(n+1)$ sets:

$$P(\underbrace{E_1 \cup \dots \cup E_n}_{\mathcal{F}} \cup E_{n+1}) \stackrel{\text{thm.}}{=} P(\mathcal{F}) + P(E_{n+1}) - P(\mathcal{F} \cap E_{n+1}) = \textcircled{*}$$

$$P(\mathcal{F}) = P(E_1 \cup \dots \cup E_n) \stackrel{\text{by our assumption}}{=} \sum_{k=1}^n \sum_{1 \leq i_1 < \dots < i_k \leq n} (-1)^{k+1} P(E_{i_1} \cap \dots \cap E_{i_k})$$

$$P(\mathcal{F} \cap E_{n+1}) = P((E_1 \cap E_{n+1}) \cup \dots \cup (E_n \cap E_{n+1})) \stackrel{\text{by our assumption}}{=} \sum_{k=1}^n \sum_{1 \leq i_1 < \dots < i_k \leq n} (-1)^{k+1} P(E_{i_1} \cap \dots \cap E_{i_k} \cap E_{n+1})$$

$$\textcircled{*} = \sum_{k=1}^{n+1} \sum_{1 \leq i_1 < \dots < i_k \leq n+1} (-1)^{k+1} P(E_{i_1} \cap \dots \cap E_{i_k})$$

□

11. Uniform distribution

Let $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$.

def. We call $\{\omega_1\}, \dots, \{\omega_N\}$ elemental events.

def. We say that $P: \mathcal{F} \rightarrow \mathbb{R}$ is uniform/uniformly distributed if $\forall i \in \{1, \dots, N\}: P(\{\omega_i\}) = 1/N$.

It follows that for all $E \in \mathcal{F}$ we have

$$P(E) = \frac{|E|}{|\Omega|} = \frac{\#\{\text{"good cases"}\}}{\#\{\text{"all cases"}\}}, \text{ since}$$

$$E = \{\omega_{i_1}, \dots, \omega_{i_k}\} \Rightarrow P(E) = \sum_{j=1}^k P(\{\omega_{i_j}\}).$$

Examples:

1/ Toss 2 coins! What is the probability that there is a head?

$$\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$$

$$E = \{(H, H), (H, T), (T, H)\}$$

$$\Rightarrow P(\text{there is a head}) = P(E) = \frac{|E|}{|\Omega|} = \frac{3}{4}$$

2/ Toss 2 dice! Which event has the larger probability?
"the sum is 10" or "the sum is 9"?

Ω	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

How can we roll 10 in total?

$$\{\text{"the sum is 10"}\} = E_1 = \{(6, 4), (4, 6), (5, 5)\}$$

How can we roll 9 in total?

$$\{\text{"the sum is 9"}\} = E_2 = \{(6, 3), (3, 6), (5, 4), (4, 5)\}$$

$$P(E_1) = \frac{3}{36} < P(E_2) = \frac{4}{36}, \text{ since } |\Omega| = 36$$

3/ There are 6 red and 5 blue balls in an urn. We pick 3 randomly. What is the probability that we picked exactly 2 red balls?

$$\Omega = \{\text{all 3 element subsets of } 5+6 \text{ balls}\}, |\Omega| = \binom{11}{3}$$

How many subsets have exactly 2 red balls?

pick two from the reds $\binom{6}{2} \cdot \binom{5}{1}$ pick the remaining one from the blues

$$\Rightarrow P(\text{exactly two reds}) = \frac{\binom{6}{2} \binom{5}{1}}{\binom{11}{3}} = \frac{\frac{6 \cdot 5}{2} \cdot 5}{\frac{11 \cdot 10 \cdot 9}{2 \cdot 3}} = \frac{5}{11}$$

4/ (Birthday paradox) In a classroom there are n many students. What is the probability that at least two students have the same birthday? (we assume $n \leq 365$)

$$\Omega = \{(i_1, \dots, i_n) : i_k \in \{1, 2, \dots, 365\} \text{ for } k \in \{1, \dots, n\}\}$$

↑ all possible birthdays the n students might have

$$A = \{(i_1, \dots, i_n) \in \Omega : \exists k \neq l \quad i_k = i_l\}$$

$$A^c = \{(i_1, \dots, i_n) \in \Omega : \forall k \neq l \quad i_k \neq i_l\}$$

It is easy to calculate $P(A^c)$!

$$P(A^c) = \frac{|A^c|}{|\Omega|} = \frac{365 \cdot 364 \cdot \dots \cdot (365 - n + 1)}{365^n} \Rightarrow P(A) = 1 - P(A^c)$$

Note: For $n=23$ $P(A) > 0.5$!

(For $n=30$ $P(A) > 0.7$, for $n=41$ $P(A) > 0.9$!)

5/ N men attend a party, and ~~everyone~~^{they all} arrive in a hat. What is the probability that ~~none~~^{none of them} picks his own hat upon departure?

$$\Omega = \{\text{all possible arrangements of the } N \text{ hats}\}, |\Omega| = N!$$

$$A_i = \{i\text{-th man picks his hat}\}, P(A_i) = |A_i|/|\Omega| = (N-1)!/N!$$

To answer the question we apply the sieve formula:

$$\{\text{none of them picks his hat}\} = A_1^c \cap A_2^c \cap \dots \cap A_N^c = \left(\bigcup_{i=1}^N A_i\right)^c$$

$$P(A_1^c \cap A_2^c \cap \dots \cap A_N^c) = P\left(\left(\bigcup_{i=1}^N A_i\right)^c\right) = 1 - P(A_1 \cup A_2 \cup \dots \cup A_N) =$$

$$\stackrel{\substack{\uparrow \\ \text{sieve formula}}}{=} 1 - \sum_{k=1}^N \sum_{1 \leq i_1 < \dots < i_k \leq N} (-1)^{k+1} \frac{P(A_{i_1} \cap \dots \cap A_{i_k})}{\frac{(N-k)!}{N!}} = 1 - \sum_{k=1}^N \binom{N}{k} (-1)^{k+1} \frac{(N-k)!}{N!} =$$

(we sum the same value for all $(i_1, \dots, i_k)$.)

$$= 1 - \sum_{k=1}^N \frac{N!}{(N-k)!k!} (-1)^{k+1} \frac{(N-k)!}{N!} = 1 + \sum_{k=1}^N \frac{(-1)^k}{k!} = \sum_{k=0}^N \frac{(-1)^k}{k!}$$

$1 = \frac{(-1)^0}{0!}$

Note: As N increases the probability converges to e^{-1} .

$$\lim_{N \rightarrow \infty} P(A_1^c \cap \dots \cap A_N^c) = \lim_{N \rightarrow \infty} \sum_{k=0}^N \frac{(-1)^k}{k!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = e^{-1}$$